16-bit ALU RTL-to-GDSII ASIC Implementation Documentation

Project Overview

This project demonstrates a complete RTL-to-GDSII ASIC implementation flow for a 16-bit Arithmetic Logic Unit (ALU) using industry-standard Cadence EDA tools.

The implementation covers the entire digital ASIC backend pipeline:

- RTL Design
- Functional Simulation
- Logic Synthesis
- Timing Constraint Definition
- MMMC Configuration
- Floorplanning
- Standard Cell Placement
- IO Pin Planning
- Clock Tree Synthesis (CTS)
- Power Distribution Network (PDN) Design
- Detailed Routing
- DRC Verification
- Connectivity Verification
- Post-Route Timing Analysis
- Final GDSII Generation

This project was implemented using:

Tool	Purpose
Cadence Xcelium	Functional simulation
Cadence Genus	RTL synthesis
Cadence Innovus	Physical design implementation
GPDK 90nm	Technology library

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1. RTL DESIGN STAGE

Objective

The objective of this stage was to design a synthesizable 16-bit ALU in Verilog HDL capable of performing arithmetic and logical operations.

RTL Files

File	Description
ALU.v	Main RTL design
tb_ALU.v	Testbench for verification

Functional Blocks Implemented

The ALU design included:

- Arithmetic operations
- Logical operations
- Shift operations
- Comparison logic
- Status flag generation
- Registered outputs

Inputs

Signal	Width	Description
clk	1	System clock
rst	1	Reset signal
A	16-bit	Operand A
B	16-bit	Operand B
opcode	4-bit	ALU operation selector
valid_in	1	Input valid signal

Outputs

Signal	Width	Description
result	16-bit	ALU output result
carry	1	Carry flag
zero	1	Zero flag
overflow	1	Overflow flag
negative	1	Negative flag
valid_out	1	Output valid signal

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2. FUNCTIONAL SIMULATION

Objective

To verify RTL functionality before synthesis.

Tool Used

Cadence Xcelium

Simulation Goals

- Verify arithmetic operations
- Verify logical operations
- Verify flag generation
- Verify sequential behavior
- Verify reset operation
- Verify timing relationships

Outputs Generated

Output	Purpose
Waveforms	Functional verification
Simulation logs	Behavioral debugging

Output	Purpose
VCD/Wave dump	Timing visualization

Key Learnings

- Clock synchronization behavior
- Sequential logic verification
- Timing-sensitive signal validation
- Waveform debugging methodology

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3. SYNTHESIS STAGE

Objective

Convert RTL design into a gate-level netlist using standard cells from the GPDK90 library.

Tool Used

Cadence Genus

Initial Challenges Faced

Library Compatibility Issues

Initially, the flow attempted to use GPDK45 libraries:

- gsclib045_v3.5
- slow.lib

This resulted in:

- Liberty parsing failures
- Missing axis definitions
- Unsupported template errors

Error encountered:

```
Improper look-up table template.
Missing axis name in template definition 'energy_template_7x1'
```

Resolution

The implementation migrated to:

GPDK90

using:

```
slow_vdd1v0_basicCells.lib
```

This resolved:

- Liberty compatibility issues
- Timing model loading issues
- Technology mapping errors

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4. TIMING CONSTRAINTS

Constraint File

```
constraint.sdc
```

Constraints Applied

Clock Definition

System clock constraints were applied using:

- create_clock
- clock uncertainty
- input delay
- output delay

Initial Timing Problems

The first synthesis attempts produced:

- unconstrained paths
- missing clock waveforms
- missing IO transitions
- missing output loads

Warnings included:

- Sequential clock pins without waveform
- Inputs without external driver/transition
- Outputs without external load

Constraint Fixes

The SDC file was updated to include:

- Proper clock definitions
- Input delays
- Output delays
- Clock uncertainty
- IO loading assumptions

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5. SYNTHESIS RESULTS

Final Synthesis Metrics

Area Report

Metric	Value
Cell Count	2657
Cell Area	5130.940
Total Area	5130.940

Power Report

Final Power Summary

Category	Total Power
Register Power	2.44778e-05 W
Logic Power	2.18199e-04 W
Clock Power	1.67891e-06 W
Total Power	2.44355e-04 W

Power Distribution

Component	Percentage
Logic	89.30%
Registers	10.02%
Clock	0.69%

Synthesis Insights

- Logic power dominated total consumption
- Clock network power remained low
- Optimized synthesis reduced overall area significantly

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6. MMMC CONFIGURATION

Objective

Enable realistic timing analysis across multiple operating conditions.

Configuration Used

- Setup View
- Hold View
- OCV Analysis

- SI-aware timing

Challenges Encountered

Initially:

- restoreDesign failures occurred
- MMMC configuration incomplete
- set_analysis_view missing
- incorrect SDC references

Resolution

The MMMC setup was rebuilt using:

- corrected SDC paths
- valid analysis views
- proper setup/hold configuration

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7. FLOORPLANNING

Objective

Define die dimensions, core region, utilization, and routing space.

Floorplan Dimensions

Die Area

```
{0.0 0.0 126.0 125.78}
```

Core Area

```
{20.0 20.14 106.0 105.64}
```

Floorplanning Goals

- Maintain routability
- Allow CTS flexibility
- Reserve IO regions
- Support future PDN insertion

Congestion Results

Congestion Overflow

```
Overflow: 0 = 0 (0.00% H) + 0 (0.00% V)
```

This indicated:

- excellent routability
- healthy floorplan spacing
- low routing pressure

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8. IO PIN PLANNING

Objective

Organize physical IO placement for clean routing.

Pin Distribution Strategy

Side	Signals
Left	A[15:0], B[15:0]
Right	result[15:0]
Top	clk, rst, opcode, valid_in
Bottom	carry, zero, overflow, negative, valid_out

Metal Layers Used

Signal Type	Layer
Data buses	Metal4
Control signals	Metal3
Status outputs	Metal3

Insights

This organization improved:

- routing cleanliness
- signal grouping
- bus routing efficiency
- congestion reduction

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9. STANDARD CELL PLACEMENT

Objective

Place synthesized standard cells physically inside the core area.

Placement Goals

- minimize wirelength
- reduce congestion
- improve timing
- optimize placement density

Placement Density

Final density reached:

```
95.479%
```

This was considered:

- aggressive
- high-utilization
- timing-sensitive

But still routable.

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10. CLOCK TREE SYNTHESIS (CTS)

Objective

Build a balanced clock distribution network.

CTS Tasks Performed

- Clock buffer insertion
- Clock skew balancing
- Clock slew optimization
- Clock latency optimization

CTS Results

Pre-CTS Timing

Metric	Value
WNS	-7.890 ns
TNS	-124.795 ns

Post-CTS Timing

Metric	Value
WNS	-1.467 ns
TNS	-20.523 ns

CTS Improvement

CTS significantly improved:

- setup slack
- total negative slack
- clock balancing
- timing closure quality

CTS Observations

Interestingly:

- no additional clock buffers were inserted
- no inverters inserted

This was acceptable because:

- design size was small
- clock fanout remained manageable
- clock distances were short

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11. POWER DISTRIBUTION NETWORK (PDN)

Objective

Create realistic power delivery infrastructure.

Initial Problem

The first routed layout lacked:

- power rings
- power stripes
- PG connectivity

Warnings included:

```
Global net connect rules have not been created
```

Solution Implemented

Global Nets Created

- VDD
- VSS

Global Connections

Applied using:

- globalNetConnect
- applyGlobalNets

Power Rings Added

Ring Configuration

Property	Value
Horizontal Layer	Metal6
Vertical Layer	Metal5
Width	2
Spacing	1
Offset	2

Stripe Configuration

Property	Value
Layer	Metal4
Direction	Vertical
Width	1
Spacing	1
Stripe Distance	20

PDN Results

Generated Structures

Element	Count
Ring wires	8
Ring vias	8
PG vias	1020
Followpin connections	51

Significance

The PDN implementation enabled:

- realistic power delivery
- standard cell rail connection
- power integrity structure
- manufacturable PG routing

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12. DETAILED ROUTING

Objective

Perform final signal routing across all nets.

Routing Tasks

- global routing
- detailed routing
- via generation
- congestion optimization
- antenna fixing

Routing Results

Total Wire Length

```
41206 um
```

Via Statistics

Metric	Value
Total vias	20035
Multi-cut via ratio	93.6%

Significance

High multi-cut via usage improved:

- reliability
- electromigration robustness
- manufacturability

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13. DRC VERIFICATION

Objective

Ensure layout geometry obeys manufacturing rules.

Initial DRC Issues

Initial PDN integration introduced:

- Metal1 violations
- spacing violations
- short violations
- end-of-line spacing issues

Total initial violations:

```
154 violations
```

Resolution

After re-routing and optimization:

Final DRC Result

```
Verification Complete : 0 Viols.
```

Significance

This indicated:

- manufacturing-clean layout
- valid geometric implementation
- successful routing cleanup

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14. CONNECTIVITY VERIFICATION

Objective

Verify all nets are fully connected.

Final Connectivity Result

```
Verification Complete : 0 Viols. 0 Wrngs.
```

Significance

This confirmed:

- no opens

- no shorts
- valid PG connectivity
- complete signal routing

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15. POST-ROUTE TIMING ANALYSIS

Objective

Perform timing analysis using:

- routed parasitics
- SI-aware delays
- OCV timing
- CPPR correction

Final Post-Route Results

Metric	Value
WNS	-1.864 ns
TNS	-25.143 ns
Violating Paths	17

Interpretation

Although setup violations remained:

- physical implementation was fully successful
- timing was significantly improved from initial synthesis
- routed parasitic realism increased timing accuracy

Important Achievement

Despite remaining timing violations:

- routing completed successfully
- DRC reached zero
- connectivity reached zero

- PDN integrated successfully

This made the design fully suitable as an educational ASIC backend implementation project.

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16. FINAL GDSII GENERATION

Objective

Export final manufacturable layout database.

Final Artifact Generated

```
final_ALU_PDN.gds
```

Significance

The GDSII file represents:

- final physical layout
- routed metal geometry
- transistor-level implementation
- manufacturable chip database

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17. FINAL PROJECT ACHIEVEMENTS

Successfully Completed

Stage	Status
RTL Design	Completed
Functional Simulation	Completed
Synthesis	Completed
MMMC Configuration	Completed
Floorplanning	Completed
IO Planning	Completed

Stage	Status
Placement	Completed
CTS	Completed
PDN Design	Completed
Detailed Routing	Completed
DRC Verification	Completed
Connectivity Verification	Completed
Post-Route Timing	Completed
GDSII Export	Completed

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18. KEY LEARNINGS

Technical Skills Learned

RTL Design

- synthesizable Verilog design
- sequential logic implementation
- status flag generation

Timing Constraints

- SDC syntax
- clock definition
- IO timing modeling
- uncertainty handling

Synthesis

- library mapping
- optimization goals
- power/area tradeoffs
- timing debugging

Physical Design

- floorplanning
- placement optimization
- congestion analysis
- IO planning

CTS

- skew balancing
- clock optimization
- latency management

PDN Engineering

- power rings
- stripes
- global nets
- followpin routing

Routing

- signal routing
- via optimization
- DRC debugging
- connectivity analysis

Signoff Verification

- DRC verification
- connectivity verification
- post-route timing
- GDS generation

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19. FINAL PROJECT SUMMARY

This project successfully demonstrated a complete end-to-end RTL-to-GDSII ASIC implementation flow for a 16-bit ALU using Cadence industry-standard EDA tools.

The implementation achieved:

- clean DRC results
- clean connectivity verification
- successful CTS
- realistic PDN implementation
- routed manufacturable layout
- final GDSII generation

The project provided practical exposure to:

- digital ASIC backend methodology
- physical implementation challenges
- timing optimization
- PDN engineering
- routing verification
- signoff-quality analysis

This implementation represents a complete educational ASIC backend project covering both frontend and backend VLSI design methodologies.

Final Artifacts

Artifact	Description
ALU.v	RTL design
tb_ALU.v	Testbench
constraint.sdc	Timing constraints
ALU_16_synth.v	Synthesized netlist
final_ALU.v	Final routed netlist
final_ALU_PDN.gds	Final physical layout
final_ALU.spef	Extracted parasitics
timing reports	Timing analysis
power reports	Power analysis
area reports	Area analysis
Innovus DB	Full physical database

End of Documentation